PART 1 · VirtualBox — Install & Configure Ubuntu Server VM

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■ PART 1 — VIRTUALBOX: INSTALL UBUNTU SERVER VM

■ VirtualBox lets you run a complete Ubuntu Server inside a window on your laptop or desktop PC. It behaves exactly like a real physical server — all CLI commands, Samba, DNS, and backup scripts work identically. Best of all, it is completely free.

Download VirtualBox and Ubuntu Server ISO

What to Download	URL	File Size
Oracle VirtualBox 7.x	virtualbox.org/wiki/Downloads	~110 MB
VirtualBox Extension Pack	Same page — 'All platforms'	~15 MB
Ubuntu Server 24.04.2 LTS	ubuntu.com/download/server	~2.6 GB

After downloading: Install VirtualBox first, then install the Extension Pack (double-click the .vbox-extpack file). Keep the Ubuntu ISO file — you will point VirtualBox to it during VM creation.

■ SCREENSHOT SS-01 | VirtualBox Manager — empty, freshly installed
Where: Open VirtualBox after installing
Shows your starting point

Create the Ubuntu Server Virtual Machine

In VirtualBox Manager click **New**. Enter these exact settings:

Setting	Value to Enter	Why
Name	school-server	Identifies this VM
Machine Folder	Leave default	Where VM files are saved
ISO Image	Browse → select ubuntu-24.04-server.iso	OS installer
Skip Unattended	■ Tick this checkbox	We install manually for control
Base Memory	2048 MB (2 GB)	Minimum; 4096 MB if you have it
Processors	2 CPUs	Adequate for a lab server
Hard Disk	Create a virtual hard disk	
Disk Size	25 GB (VDI, Dynamically allocated)	Enough for all labs
Hard Disk Type	VDI (VirtualBox Disk Image)	Default, fine for labs

Network Adapter Settings (critical — do this BEFORE first boot):

Go to VM Settings → Network:

Adapter	Attached To	Purpose
Adapter 1	NAT	Internet access — to download packages (apt install)
Adapter 2	Internal Network → Name: schoolnet	Simulates the school LAN — for connecting client VMs

■ SCREENSHOT SS-02 | VM Settings — Network tab showing both adapters configured

Where: VM → Settings → Network

Prove adapter 1 = NAT, adapter 2 = Internal Network: schoolnet

1.3 Install Ubuntu Server 24.04 LTS Inside the VM

Start the VM (click Start ■). The Ubuntu installer boots from the ISO automatically.

Installer Screen	What to Select / Type
Language	English
Keyboard layout	Your country layout (e.g. English UK / US)
Type of install	Ubuntu Server (not minimised)
Network — enp0s3 (NAT)	Leave as DHCP — this gives internet access
Network — enp0s8 (Internal)	Manual → IP: 192.168.60.10 / 24 Gateway: (blank)
Proxy	Leave blank → Done
Mirror	Leave default → Done
Storage	Use entire disk → Set up LVM → Done → Continue
Your name	IT Admin
Server name	school-server
Username	itadmin
Password	Choose a strong password (write it down!)
OpenSSH server	■ TICK THIS — Install OpenSSH server
Featured snaps	Skip — press Done without selecting anything
Installation complete	Click Reboot Now → press Enter when asked to remove media

■ SCREENSHOT SS-03 | Ubuntu installer — Profile setup screen
Where: During install, the 'Profile setup' screen with server name + username
Shows school-server hostname and itadmin username

■ SCREENSHOT SS-04 | Ubuntu Server login prompt after reboot
Where: After reboot — the black console login screen
Type itadmin and password to log in

1.4 Post-Install: Update System and Verify Network

UBUNTU VM — FIRST LOGIN & POST-INSTALL UPDATE

```
# Log in as itadmin
# First — update all packages
```

```
sudo apt update && sudo apt upgrade -y

# Check both network interfaces are up
ip addr show
# enp0s3: should have a 10.0.x.x address (NAT / internet)
# enp0s8: should have 192.168.60.10 (Internal / school LAN)

# Confirm hostname
hostname
# Output: school-server

# Check internet works (downloads via NAT adapter)
ping -c 3 google.com

# Install useful tools
sudo apt install -y net-tools curl wget tree htop
```

■ SCREENSHOT SS-05 | ip addr show — showing both network adapters with IPs

Where: After running: ip addr show

Both enp0s3 (NAT) and enp0s8 (192.168.60.10) should be visible

■■ PART 2 — CONFIGURE THE SERVER (CLI COMMANDS IN VM)

2.1 Set Static IP Address with Netplan

NETPLAN — STATIC IP FOR SCHOOL LAN ADAPTER

```
# Open the netplan config file
sudo nano /etc/netplan/00-installer-config.yaml

# DELETE everything and paste EXACTLY this:
network:
version: 2
renderer: networkd
ethernets:
enp0s3:
dhcp4: true # NAT adapter — keeps internet access
enp0s8:
dhcp4: no
addresses:
- 192.168.60.10/24 # Static IP on school LAN
nameservers:
addresses: [8.8.8.8, 1.1.1.1]

# Save: Ctrl+O then Enter, then Ctrl+X to exit nano

# Apply the config
sudo netplan apply

# Verify — should show 192.168.60.10 on enp0s8
ip addr show enp0s8
```

■ SCREENSHOT SS-06 | Netplan YAML file open in nano editor
Where: During: sudo nano /etc/netplan/00-installer-config.yaml
Show the YAML content with the static IP visible

2.2 Install Samba and Create Department Shares

SAMBA STEP 1 — CREATE SHARE DIRECTORIES

```
# Install Samba
sudo apt install samba samba-common-bin -y

# Create department directories
sudo mkdir -p /shares/admin /shares/academic /shares/students
sudo mkdir -p /shares/library /shares/finance /shares/it

# Create Linux groups for each department
sudo groupadd school-admin
sudo groupadd school-academic
sudo groupadd school-students
sudo groupadd school-library
sudo groupadd school-finance
sudo groupadd school-it

# Set ownership and permissions
sudo chown -R root:school-admin /shares/admin
sudo chown -R root:school-academic /shares/academic
sudo chown -R root:school-students /shares/students
sudo chown -R root:school-library /shares/library
sudo chown -R root:school-finance /shares/finance
sudo chown -R root:school-it /shares/it

sudo chmod -R 2770 /shares/admin
sudo chmod -R 2770 /shares/academic
sudo chmod -R 2775 /shares/students
sudo chmod -R 2770 /shares/library
sudo chmod -R 2770 /shares/finance
sudo chmod -R 2700 /shares/it

# Verify structure
ls -la /shares/
```

■ SCREENSHOT SS-07 | ls -la /shares/ — showing all 6 department folders

Where: After running: ls -la /shares/

All folders visible with correct group ownership

SAMBA STEP 2 — smb.conf CONFIGURATION

```
# Back up original Samba config
sudo cp /etc/samba/smb.conf /etc/samba/smb.conf.backup

# Open config file
```

```
sudo nano /etc/samba/smb.conf

# ADD these lines at the VERY END of the file:

[global]
workgroup = SCHOOLNET
server string = School File Server
security = user
map to guest = bad user
log file = /var/log/samba/log.%m
max log size = 50

[Admin]
path = /shares/admin
valid users = @school-admin
read only = no
browseable = yes
create mask = 0660
directory mask = 0770
comment = Administration Department

[Academic]
path = /shares/academic
valid users = @school-academic @school-admin
read only = no
browseable = yes
create mask = 0660
directory mask = 0770
comment = Teachers and Academic Staff

[Students]
path = /shares/students
valid users = @school-students @school-academic
read only = yes
browseable = yes
comment = Student Resources - Read Only

[Library]
path = /shares/library
```

```
valid users = @school-library @school-academic
read only = no
browseable = yes
comment = Library Resources

[Finance]
path = /shares/finance
valid users = @school-finance
read only = no
browseable = no
comment = Finance - Restricted Access

[IT]
path = /shares/it
valid users = @school-it
read only = no
browseable = no
comment = IT Administration Only

# Save: Ctrl+O, Enter, Ctrl+X

# Test the config — should say 'Loaded services file OK'
testparm

# Enable and start Samba
sudo systemctl enable smbd nmbd
sudo systemctl start smbd nmbd
sudo systemctl status smbd
```

■ SCREENSHOT SS-08 | testparm output — 'Loaded services file OK'

Where: After running: testparm

No errors = config is correct. All 6 share sections should appear

■ SCREENSHOT SS-09 | systemctl status smbd — showing active (running)

Where: After running: sudo systemctl status smbd

Green 'active (running)' dot = Samba is working

2.3 Create Users and Add to Department Groups

USERS — CREATE DEPARTMENT ACCOUNTS

```
# Create teacher account
sudo useradd -m -s /bin/bash mrs.okonkwo
sudo usermod -aG school-academic mrs.okonkwo
sudo smbpasswd -a mrs.okonkwo
# Enter password: Teacher@123 (type twice)

# Create another teacher
sudo useradd -m -s /bin/bash mr.adeyemi
sudo usermod -aG school-academic mr.adeyemi
sudo smbpasswd -a mr.adeyemi

# Create admin staff
sudo useradd -m -s /bin/bash registrar
sudo usermod -aG school-admin registrar
sudo smbpasswd -a registrar

# Create student account
sudo useradd -m -s /bin/bash student.john
sudo usermod -aG school-students student.john
sudo smbpasswd -a student.john

# Create finance user
sudo useradd -m -s /bin/bash bursar
sudo usermod -aG school-finance bursar
sudo smbpasswd -a bursar

# List all Samba users
sudo pdbedit -L

# Verify group membership
groups mrs.okonkwo
# Output: mrs.okonkwo : mrs.okonkwo school-academic
```

■ SCREENSHOT SS-10 | sudo pdbedit -L — listing all created Samba users

Where: After running: sudo pdbedit -L

Should show 5 users: mrs.okonkwo, mr.adeyemi, registrar, student.john, bursar

2.4 Configure UFW Firewall

UFW FIREWALL — ALLOW REQUIRED SERVICES

```
# Enable UFW (Ubuntu's firewall)
sudo ufw enable
# Type 'y' to confirm

# Allow SSH (so you can connect remotely — important!)
sudo ufw allow OpenSSH

# Allow Samba file sharing
sudo ufw allow Samba

# Allow DNS queries (for dnsmasq)
sudo ufw allow 53/tcp
sudo ufw allow 53/udp

# Allow DHCP
sudo ufw allow 67/udp
sudo ufw allow 68/udp

# View all rules
sudo ufw status verbose
```

■ SCREENSHOT SS-11 | sudo ufw status verbose — showing all firewall rules

Where: After running: sudo ufw status verbose

Should show SSH, Samba, DNS ports as ALLOW

2.5 Set Up Automated Backup with rsync + cron

BACKUP — rsync SCRIPT + CRON SCHEDULE

```
# Create backup destination directory
sudo mkdir -p /backup

# Create the backup script
sudo nano /usr/local/bin/school-backup.sh

# Paste this content:
```

```
#!/bin/bash
DATE=$(date +%Y-%m-%d_%H-%M)
DEST="/backup/$DATE"
mkdir -p "$DEST"
rsync -avz /shares/ "$DEST/" >> /var/log/school-backup.log 2>&1
echo "===== Backup done: $DATE =====" >> /var/log/school-backup.log
find /backup -maxdepth 1 -type d -mtime +30 -exec rm -rf {} \;

# Save (Ctrl+O, Enter, Ctrl+X)

# Make it executable
sudo chmod +x /usr/local/bin/school-backup.sh

# Run it now to test
sudo /usr/local/bin/school-backup.sh

# Confirm backup was created
ls -lh /backup/

# View the log
cat /var/log/school-backup.log

# Schedule: run every night at 11 PM
sudo crontab -e
# Choose editor 1 (nano) when asked
# Add this line at the bottom:
0 23 * * * /usr/local/bin/school-backup.sh
# Save and exit

# Verify cron entry
sudo crontab -l
```

■ SCREENSHOT SS-12 | ls -lh /backup/ — showing timestamped backup folder

Where: After running: ls -lh /backup/

A folder like 2025-05-26_23-00 should appear

■ SCREENSHOT SS-13 | sudo crontab -l — showing the 11 PM cron entry

Where: After running: sudo crontab -l

The 0 23 * * * line must be visible

■ PART 3 — CISCO PACKET TRACER: SCHOOL NETWORK TOPOLOGY

Cisco Packet Tracer is a free network simulator from Cisco. You will build the complete school network visually — switches, router, VLANs, department PCs, and a server — and test it with live ping and traceroute. No real hardware needed whatsoever.

3.1 Download and Install Cisco Packet Tracer (Free)

Follow these steps to get Packet Tracer for free:

Step	Action
1	Go to: netacad.com → Click 'Sign up' → Create a free Cisco Networking Academy account
2	After login → go to: skillsforall.com → Search 'Getting Started with Cisco Packet Tracer'
3	Enrol in the free course (no payment) → This unlocks the Packet Tracer download
4	Download Packet Tracer 8.2.x for your OS (Windows / macOS / Linux)
5	Install it → Open → Log in with your Cisco NetAcad account

■ SCREENSHOT SS-PT-01 | Cisco Packet Tracer — blank workspace open
Where: After opening Packet Tracer and logging in
The empty canvas where you will build the topology

3.2 Build the School Network Topology — Devices to Place

Work from the bottom panel of Packet Tracer. Device types are shown below. Click a device type, then click on the canvas to place it.

Device	Model to Use	Quantity	Where to Find in PT	Rename To
Router	Cisco 4321 or 2911	1	Routers category	School-Router
Managed Switch	Cisco 3650-24PS	1	Switches → Catalyst	School-Switch
Server	Generic Server	1	End Devices → Server	School-Server
Admin PC	Generic PC	1	End Devices → PC	Admin-PC
Teacher PC	Generic PC	1	End Devices → PC	Teacher-PC
Student PC	Generic PC	1	End Devices → PC	Student-PC
Library PC	Generic PC	1	End Devices → PC	Library-PC
Finance PC	Generic PC	1	End Devices → PC	Finance-PC

Wireless AP	Access Point (WRT)	1	Wireless Devices	Student-WiFi
Laptop	Generic Laptop	1	End Devices → Laptop	Student-Laptop

Cable Connections (use Copper Straight-Through cable — solid line icon):

From Device	From Port	To Device	To Port
School-Router	Gig0/0/0	School-Switch	Gig1/0/1 (uplink)
School-Switch	Gig1/0/2	School-Server	FastEthernet0
School-Switch	Gig1/0/3	Admin-PC	FastEthernet0
School-Switch	Gig1/0/4	Teacher-PC	FastEthernet0
School-Switch	Gig1/0/5	Student-PC	FastEthernet0
School-Switch	Gig1/0/6	Library-PC	FastEthernet0
School-Switch	Gig1/0/7	Finance-PC	FastEthernet0
School-Switch	Gig1/0/8	Student-WiFi	Internet port
Student-WiFi	Ethernet	Student-Laptop	Wireless

■ SCREENSHOT SS-PT-02 | Full topology — all devices placed and cabled

Where: After placing all devices and drawing all cables

All cable connections should show green dots (link up)

3.3 Configure VLANs on the Managed Switch (CLI)

Click on School-Switch → CLI tab. Click on the black terminal area and press Enter. Then type the following commands:

PACKET TRACER — SWITCH — VLAN CONFIGURATION

```
! ■■ Enter privileged mode ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
enable
configure terminal

! ■■ Create VLANs ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
vlan 10
name ADMIN
vlan 20
name ACADEMIC
vlan 30
name STUDENTS
vlan 40
name LIBRARY
vlan 50
name FINANCE
vlan 60
name IT-SERVERS
exit

! ■■ Assign ports to VLANs ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
! Admin-PC on port Gig1/0/3 → VLAN 10
interface GigabitEthernet1/0/3
switchport mode access
switchport access vlan 10
no shutdown

! Teacher-PC on port Gig1/0/4 → VLAN 20
interface GigabitEthernet1/0/4
switchport mode access
switchport access vlan 20
no shutdown
```

```
! Student-PC on port Gig1/0/5 → VLAN 30
interface GigabitEthernet1/0/5
switchport mode access
switchport access vlan 30
no shutdown

! Library-PC on port Gig1/0/6 → VLAN 40
interface GigabitEthernet1/0/6
switchport mode access
switchport access vlan 40
no shutdown

! Finance-PC on port Gig1/0/7 → VLAN 50
interface GigabitEthernet1/0/7
switchport mode access
switchport access vlan 50
no shutdown

! Server on port Gig1/0/2 → VLAN 60
interface GigabitEthernet1/0/2
switchport mode access
switchport access vlan 60
no shutdown

! Uplink to Router — Trunk (carries all VLANs)
interface GigabitEthernet1/0/1
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60
no shutdown

exit

! ■■ Save configuration ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
write memory

! ■■ Verify ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
show vlan brief
```

■ SCREENSHOT SS-PT-03 | Switch CLI — show vlan brief output

Where: Switch → CLI → after: show vlan brief

All 6 VLANs (10,20,30,40,50,60) listed with correct port assignments

3.4 Configure the Router — Inter-VLAN Routing

Click on School-Router → CLI tab. This sets up sub-interfaces (Router-on-a-Stick) so traffic can flow between VLANs when permitted:

PACKET TRACER — ROUTER — INTER-VLAN ROUTING

```
enable
configure terminal

! Main interface to switch (no IP — sub-interfaces carry traffic)
interface GigabitEthernet0/0/0
no shutdown

! Sub-interface for VLAN 10 (Admin) — 192.168.10.0/24
interface GigabitEthernet0/0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
no shutdown

! Sub-interface for VLAN 20 (Academic) — 192.168.20.0/24
interface GigabitEthernet0/0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
no shutdown

! Sub-interface for VLAN 30 (Students) — 192.168.30.0/24
interface GigabitEthernet0/0/0.30
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
no shutdown

! Sub-interface for VLAN 40 (Library) — 192.168.40.0/24
interface GigabitEthernet0/0/0.40
encapsulation dot1Q 40
ip address 192.168.40.1 255.255.255.0
no shutdown

! Sub-interface for VLAN 50 (Finance) — 192.168.50.0/24
```

```
interface GigabitEthernet0/0/0.50
encapsulation dot1Q 50
ip address 192.168.50.1 255.255.255.0
no shutdown

! Sub-interface for VLAN 60 (IT/Server) — 192.168.60.0/24
interface GigabitEthernet0/0/0.60
encapsulation dot1Q 60
ip address 192.168.60.1 255.255.255.0
no shutdown

exit
write memory

! Verify sub-interfaces
show ip interface brief
```

■ SCREENSHOT SS-PT-04 | Router CLI — show ip interface brief

Where: Router → CLI → after: show ip interface brief

All 6 sub-interfaces (Gig0/0/0.10 through .60) should show 'up/up'

3.5 Configure IP Addresses on Department PCs

For each PC: click it → Desktop tab → IP Configuration. Enter these values:

PC Name	IP Address	Subnet Mask	Default Gateway	VLAN
Admin-PC	192.168.10.10	255.255.255.0	192.168.10.1	VLAN 10
Teacher-PC	192.168.20.10	255.255.255.0	192.168.20.1	VLAN 20
Student-PC	192.168.30.10	255.255.255.0	192.168.30.1	VLAN 30
Library-PC	192.168.40.10	255.255.255.0	192.168.40.1	VLAN 40
Finance-PC	192.168.50.10	255.255.255.0	192.168.50.1	VLAN 50
School-Server	192.168.60.10	255.255.255.0	192.168.60.1	VLAN 60

■ SCREENSHOT SS-PT-05 | Admin-PC IP Configuration screen showing 192.168.10.10

Where: Admin-PC → Desktop → IP Configuration

Static IP, subnet, and gateway fields all filled in

3.6 Test Connectivity — Ping Between Departments

Use the Command Prompt on each PC (PC → Desktop → Command Prompt) to test. Also use the Packet Tracer 'Add Simple PDU' tool (envelope icon) for visual ping.

PACKET TRACER — PING TEST + ACCESS CONTROL LIST

```
! ■■ ON Admin-PC Command Prompt ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
ping 192.168.60.10
! Expected: Reply from 192.168.60.10 — Admin can reach server ■

ping 192.168.20.10
! Expected: Reply — Admin can reach Academic VLAN ■

! ■■ ON Student-PC Command Prompt ■■■■■■■■■■■■■■■■■■■■■■■■■■■■
ping 192.168.60.10
! Expected: Reply — Students can reach server to access shares ■

ping 192.168.50.10
! Expected: Reply — BUT we will block this with ACL (see below)
```

```
! ■■ ACL: Block Students from Finance ■■■■■■■■■■■■■■■■■■■■■■■■■
! On School-Router CLI:
enable
configure terminal
ip access-list extended BLOCK_STUDENTS_FINANCE
deny ip 192.168.30.0 0.0.0.255 192.168.50.0 0.0.0.255
permit ip any any
interface GigabitEthernet0/0/0.30
ip access-group BLOCK_STUDENTS_FINANCE in
exit
write memory

! ■■ Re-test after ACL ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
! ON Student-PC:
ping 192.168.50.10
! Expected: Request timed out — ACCESS BLOCKED ■

! ■■ Traceroute from Teacher-PC to Server ■■■■■■■■■■■■■■■■■■■■■
tracert 192.168.60.10
! Shows the path: Teacher → Router → Server
```

■ SCREENSHOT SS-PT-06 | Admin-PC ping to server — 4 successful replies

Where: Admin-PC → Desktop → Command Prompt → ping 192.168.60.10

4 green reply lines = connectivity working

■ SCREENSHOT SS-PT-07 | Student-PC ping to Finance — 'Request timed out'

Where: Student-PC → Command Prompt → ping 192.168.50.10 (after ACL)

This is your SECURITY DEMO — access blocked by ACL ■

■ SCREENSHOT SS-PT-08 | Full topology view with all green link lights

Where: Packet Tracer canvas — zoom out to see all devices

All cables should show green dots at both ends

3.7 Configure Server Services in Packet Tracer

Click on School-Server → Services tab. Packet Tracer's generic server has built-in services you can enable to simulate a real server:

Service	Turn On?	Settings to Configure
HTTP	■ ON	Default page shows 'School Server' — edit the HTML if you like
DHCP	■ ON	Pool: serverPool → Start: 192.168.60.100, Max: 50, Gateway: 192.168.60.1
DNS	■ ON	Add record: school-server.local → 192.168.60.10 (Type A)
FTP	■ ON	Add user: itadmin / password: Admin@123 / Permission: Read+Write
TFTP	OFF	Not needed for this lab
AAA	OFF	Not needed for this lab

■ SCREENSHOT SS-PT-09 | Server Services tab — HTTP, DHCP, DNS, FTP all ON

Where: School-Server → Services tab

Show all 4 services toggled ON with green indicators

■ SCREENSHOT SS-PT-10 | DNS service — A record for school-server.local

Where: School-Server → Services → DNS

Record: school-server.local → 192.168.60.10 visible in the table

■ PART 4 — COMPLETE SCREENSHOT MASTER CHECKLIST

VirtualBox / Ubuntu Server Screenshots (20 shots)

#	Screenshot	Command / Where	Tick When Done
VB-01	VirtualBox Manager — VM listed	VirtualBox main window	■
VB-02	VM Settings — Network adapters	VM → Settings → Network	■
VB-03	Ubuntu installer — Profile setup screen	During OS install	■
VB-04	Ubuntu first login prompt (black screen)	After reboot	■
VB-05	ip addr show — both adapters with IPs	sudo ip addr show	■
VB-06	Netplan YAML file in nano	sudo nano /etc/netplan/...	■
VB-07	ls -la /shares/ — 6 department folders	ls -la /shares/	■
VB-08	smb.conf open in nano — share sections visible	sudo nano /etc/samba/smb.conf	■
VB-09	testparm — 'Loaded services file OK'	testparm	■
VB-10	systemctl status smbd — green active	sudo systemctl status smbd	■
VB-11	sudo pdbedit -L — all 5 users listed	sudo pdbedit -L	■
VB-12	sudo ufw status verbose	sudo ufw status verbose	■
VB-13	Backup script open in nano	sudo nano /usr/local/bin/school-backup.sh	■
VB-14	ls -lh /backup/ — backup folder exists	ls -lh /backup/	■
VB-15	cat /var/log/school-backup.log	cat /var/log/school-backup.log	■
VB-16	sudo crontab -l — showing 0 23 * * *	sudo crontab -l	■
VB-17	smbstatus — showing active connections	sudo smbstatus	■
VB-18	df -h /shares — disk usage	df -h /shares	■
VB-19	Windows File Explorer — mapped Z: drive showing folders	Windows client PC	■
VB-20	Access Denied when student tries Finance share	Windows client PC	■

Cisco Packet Tracer Screenshots (15 shots)

#	Screenshot	Where in Packet Tracer	Tick
PT-01	Empty PT workspace — freshly opened	After launch	■
PT-02	All devices placed — before cabling	Canvas with devices	■
PT-03	Full topology — all cables green	Canvas — zoomed out	■
PT-04	Switch CLI — show vlan brief	Switch → CLI	■
PT-05	Switch CLI — show interfaces trunk	Switch → CLI	■

PT-06	Router CLI — show ip interface brief	Router → CLI	■
PT-07	Admin-PC IP Configuration screen	Admin-PC → Desktop	■
PT-08	Admin-PC ping to server — success	Admin-PC → Command Prompt	■
PT-09	Student ping to Finance — blocked	Student-PC → Command Prompt	■
PT-10	Teacher traceroute to server	Teacher-PC → Command Prompt	■
PT-11	Server Services tab — all ON	School-Server → Services	■
PT-12	Server DNS config — A record	Server → Services → DNS	■
PT-13	Simulation mode — packet animation	Simulation tab → ping	■
PT-14	Router ACL — show access-lists	Router → CLI	■
PT-15	Final topology — labelled and tidy	Canvas — full view	■

HOW TO STRUCTURE YOUR GITHUB REPOSITORY

Recommended Folder Structure for Your GitHub Project

GITHUB — RECOMMENDED REPOSITORY STRUCTURE

```
school-server-project/
■■■ README.md ← Main project description + all quotes
■■■ docs/
■ ■■■ School_Server_Project.pdf ← PDF 1 (concept + CLI guide)
■ ■■■ School_Server_LAB_GUIDE.pdf ← PDF 2 (this guide)
■■■ virtualbox/
■ ■■■ screenshots/
■ ■ ■■■ VB-01_virtualbox_manager.png
■ ■ ■■■ VB-02_network_adapters.png
■ ■ ■■■ ... (all 20 VirtualBox shots)
■ ■■■ configs/
■ ■■■ smb.conf
■ ■■■ netplan-config.yaml
■ ■■■ school-backup.sh
■■■ packet-tracer/
■ ■■■ screenshots/
■ ■ ■■■ PT-01_blank_workspace.png
■ ■ ■■■ PT-03_full_topology.png
■ ■ ■■■ ... (all 15 PT shots)
■ ■■■ school-network.pkt ← Save your PT file and upload it
■■■ scripts/
■■■ school-backup.sh
■■■ add-user.sh
■■■ check-server-health.sh
```

README.md Opening Section Template

README.md — OPENING TEMPLATE

```
# ■ School Server Infrastructure Project

> Replacing filing cabinets and USB drives with a centralised,
> VLAN-segmented Ubuntu Server — built and demonstrated entirely
> in a free virtual lab (VirtualBox + Cisco Packet Tracer).
```

```
## ■ Tools Used
| Tool | Purpose | Cost |
|---|---|---|
| Oracle VirtualBox 7 | Ubuntu Server VM | FREE |
| Ubuntu Server 24.04 LTS | Server OS | FREE |
| Cisco Packet Tracer 8.2 | Network simulation | FREE |
| Samba | File sharing (SMB) | FREE |
| OpenLDAP | Centralised authentication | FREE |

## ■ What This Project Demonstrates
- ■ Centralised file storage for 6 departments
- ■ VLAN network segmentation (VLANs 10–60)
- ■ Role-based access control (students cannot access finance)
- ■ Automated nightly backup with rsync + cron
- ■ Full CLI configuration walkthrough
- ■ Plug-and-use guide for non-technical staff

## ■ See the Screenshots
[/virtualbox/screenshots] | [/packet-tracer/screenshots]
```

> ***"You do not need a server room to learn server administration.***
> ***You need a laptop, VirtualBox, and this guide."***
> ***— Build the lab. Run the commands. Take the screenshots. Post the project.***

COPY-PASTE CAPTIONS FOR YOUR POSTS

Reddit r/sysadmin · r/homelab · r/linuxadmin

> ***Built a complete school server infrastructure using only free tools. VirtualBox + Ubuntu Server for the CLI work, Cisco Packet Tracer for the network topology and VLAN segmentation. No physical hardware at all. Full repo with configs, screenshots and step-by-step guide linked below. [OC]***

LinkedIn

> ***■ You don't need a server room to build server skills.***

I simulated a complete school network infrastructure using:
• Oracle VirtualBox (free) → Ubuntu Server 24.04 with Samba, DNS, backup automation
• Cisco Packet Tracer (free) → 6-VLAN school network with inter-VLAN routing and ACLs

Total hardware cost: $0. Everything runs on a regular laptop.

Full project with CLI commands, screenshots, and configs on GitHub.

#Linux #NetworkAdmin #SysAdmin #EdTech #CiscoPacketTracer #Ubuntu #SchoolIT

GitHub Release Description

v1.0 — Full school server virtual lab. Includes VirtualBox Ubuntu Server config (Samba, DNS, backup), Cisco Packet Tracer .pkt topology file (6 VLANs, inter-VLAN routing, ACLs), 35 annotated screenshots, and two detailed PDF guides. Zero cost. Zero hardware required.